# EXPERIMENT 9
# LID-DRIVEN CAVITY FLOW

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## 1. Aim

To solve the two-dimensional incompressible lid-driven cavity flow using the vorticity–streamfunction formulation and examine the primary recirculation through centreline velocity profiles, streamfunction contours and the numerical convergence history.

## 2. Governing equations

For an incompressible, two-dimensional flow, the streamfunction $\psi$ is used to satisfy continuity automatically. The velocity components are:

$$u = \partial\psi/\partial y \qquad \text{and} \qquad v = -\partial\psi/\partial x$$

The scalar vorticity is defined as:

$$\omega = \partial v/\partial x - \partial u/\partial y = -\nabla^2\psi$$

The vorticity transport equation is:

$$\partial\omega/\partial t + u(\partial\omega/\partial x) + v(\partial\omega/\partial y) = \nu\nabla^2\omega$$

The streamfunction Poisson equation is:

$$\nabla^2\psi = -\omega$$

Here, u and v are velocity components, $\omega$ is vorticity, $\psi$ is streamfunction, and $\nu$ is kinematic viscosity. The Reynolds number is $Re = U_{lid} L/\nu$. For the unit square and $U_{lid} = 1$, $\nu = 1/Re$.

## 3. Computational domain and boundary conditions

The computational domain is a square cavity of length $L = 1$ and height $H = 1$. The top lid moves horizontally at a constant speed $U_{lid} = 1$. The other three walls are stationary.

- Streamfunction: $\psi = 0$ on all four walls (constant streamfunction on the closed cavity boundary).
- Top moving lid: $u = U_{lid} = 1$ and $v = 0$.
- Bottom wall: $u = 0$ and $v = 0$.
- Left and right walls: $u = 0$ and $v = 0$.
- Vorticity: wall vorticity is evaluated from the streamfunction and wall velocity using the wall-vorticity boundary formulas implemented in the solver.

The code imposes zero streamfunction on all boundaries. It calculates wall vorticity using one-sided wall relations; for example, the moving-lid relation includes the lid-speed contribution $-2U_{lid}/\Delta y$.

## 4. Numerical method

The solution uses an explicit vorticity-transport update and an iterative Successive Over-Relaxation (SOR) method for the streamfunction Poisson equation. Velocities are obtained by differentiating the streamfunction. Spatial convection derivatives are evaluated with first-order upwind differences, and the diffusion term uses a central-difference Laplacian. The wall-vorticity values are then reapplied. The solver records the RMS change in interior vorticity at each time step.

The time step is selected from the convective and diffusive restrictions in the supplied code:

$$\Delta t = \min[0.25\,CFL\,\min(\Delta x,\Delta y)/U_{lid},\ 0.05\,\min(\Delta x^2,\Delta y^2)/\nu]$$

## 5. Complete parameter set and grid used for the supplied result

| | |
|---|---|
| Run mode evidenced by result | quick |
| Grid | nx = 21, ny = 21 (21 × 21 nodes) |
| Domain | 1 × 1 square cavity |
| Grid spacing | $\Delta x = \Delta y = 1/(21-1) = 0.05$ |
| Reynolds number | Re = 100 |
| Lid speed | $U_{lid} = 1.0$ |
| Kinematic viscosity | $\nu = U_{lid}/Re = 0.01$ |
| Time step | $\Delta t = 0.01$ (reported by solver) |
| Maximum / completed steps | 10 / 10 |
| Output interval | 10 steps |
| Poisson iterations per time step | 10 |
| Convergence tolerance | $1 \times 10^{-5}$ |
| CFL parameter | 0.8 |
| SOR relaxation factor | 1.5 |
| Final vorticity-change L2 norm | $9.726355765939 \times 10^{-2}$ |
| Minimum streamfunction | $-2.360796139808 \times 10^{-2}$ |
| Primary vortex location | x = 0.55, y = 0.90 |

## 6. Numerical summary and primary vortex

| Quantity | Reported value |
|---|---|
| Reynolds number | 100 |
| Completed time steps | 10 |
| Final vorticity-change L2 norm | 0.09726356 |
| Minimum streamfunction | −0.02360796 |
| Primary vortex centre (x, y) | (0.55, 0.90) |

The minimum streamfunction is negative under the sign convention used in the solver. The location of this minimum is reported as the primary-vortex position. In the supplied quick result it is at (0.55, 0.90), close to the upper region of the cavity. This location should be treated as a preliminary estimate because the run has only ten steps and a coarse grid.

## 7. Centreline velocity profiles

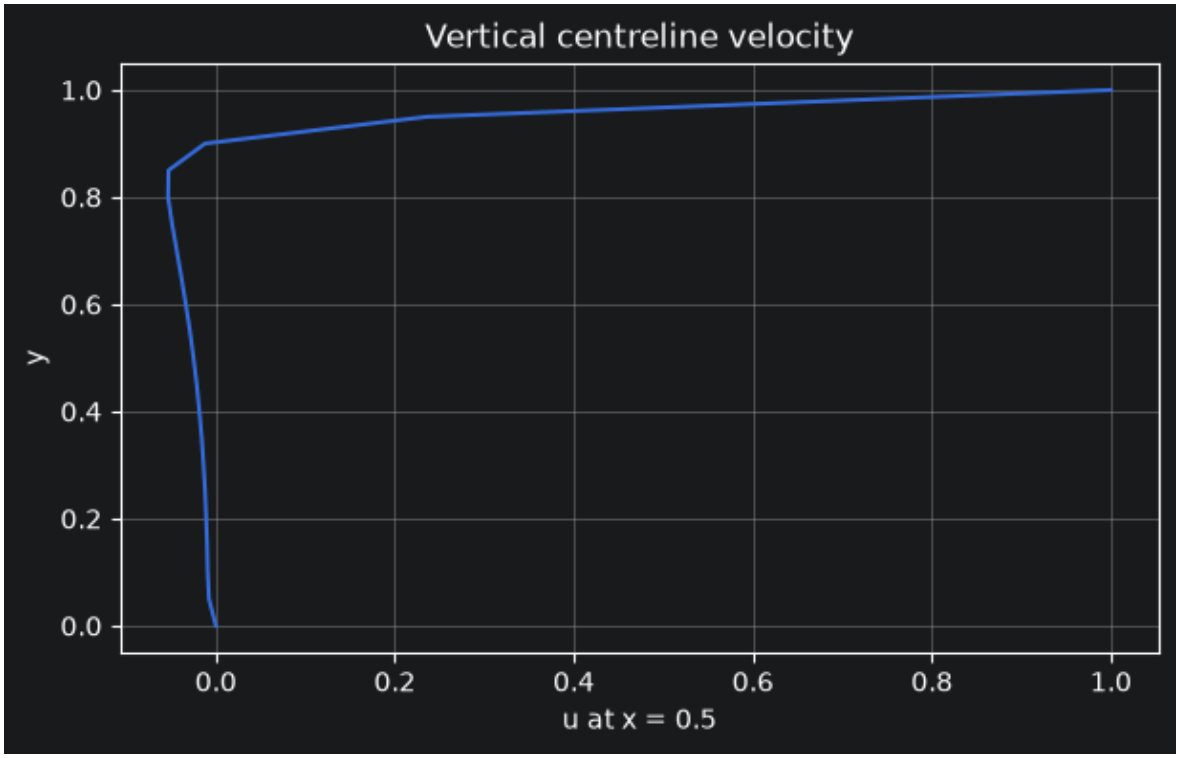



Figure 1. Vertical centreline horizontal velocity, u, plotted against y at x = 0.5.

The vertical centreline profile shows the horizontal velocity changing from approximately zero near the bottom wall to the moving-lid velocity near the top. The profile has a small negative-velocity region in the upper part of the cavity before increasing towards the lid speed. This reversal is associated with the recirculating motion generated by the moving lid.

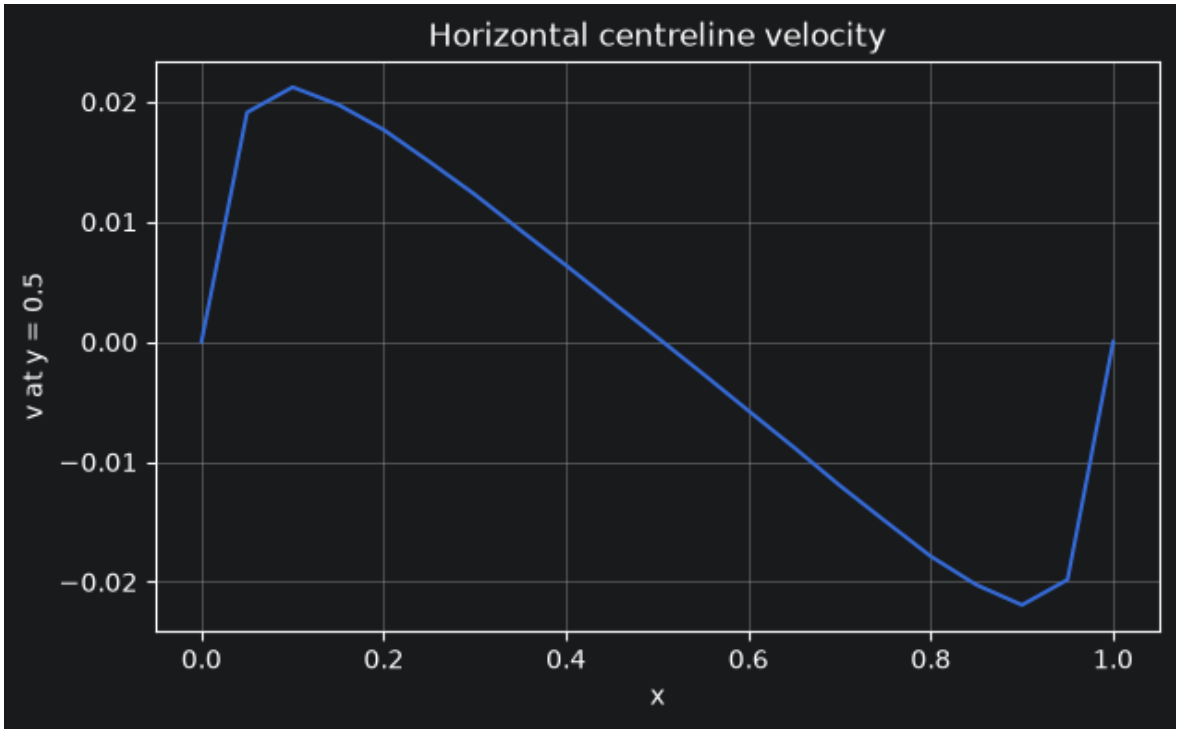



Figure 2. Horizontal centreline vertical velocity, v, plotted against x at y = 0.5.

The horizontal centreline profile is positive over the left portion of the cavity and negative over the right portion, with a sign change near the centre. This pattern indicates the return-flow portion of the main recirculation cell. The magnitudes shown are small in this short, coarse-grid calculation.

## 8. Convergence history

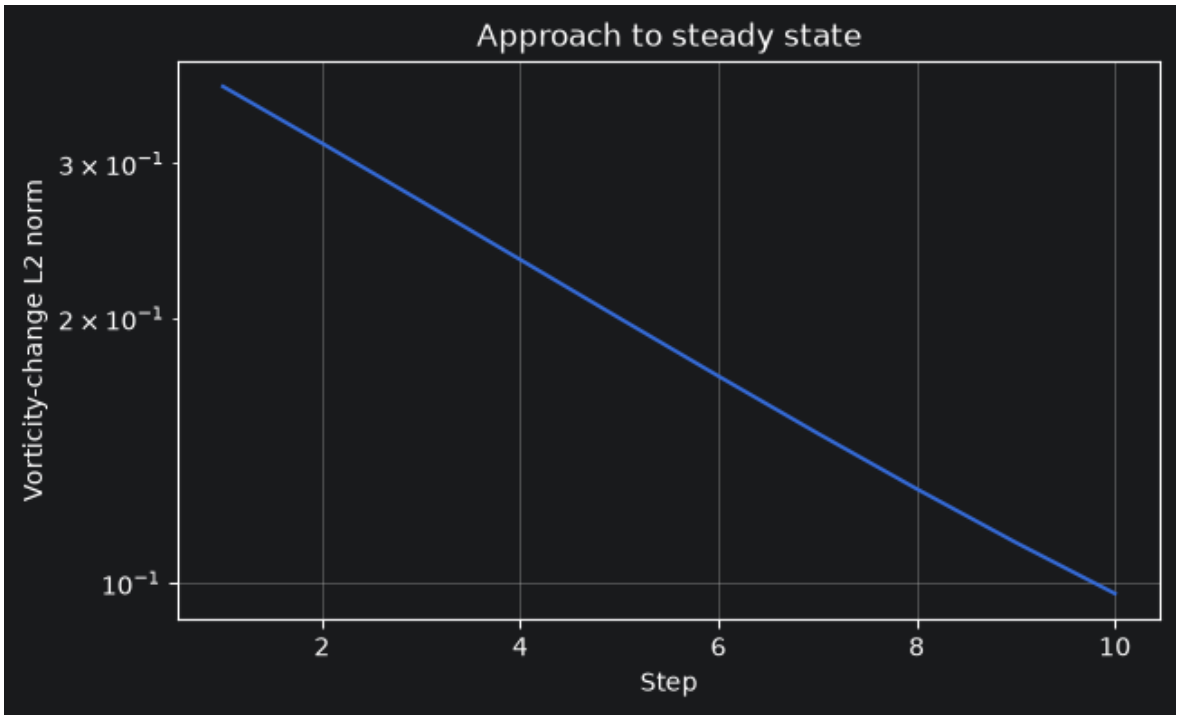



Figure 3. Vorticity-change L2 norm versus iteration step.

The plotted vorticity-change L2 norm decreases from approximately 0.36 at the first step to approximately 0.097 at step 10. The decreasing trend indicates that the solution is moving towards a steady state. However, the final value is not close to the specified tolerance of $1 \times 10^{-5}$. Therefore, the supplied run has not converged to the solver tolerance.

## 9. Streamfunction field visualization

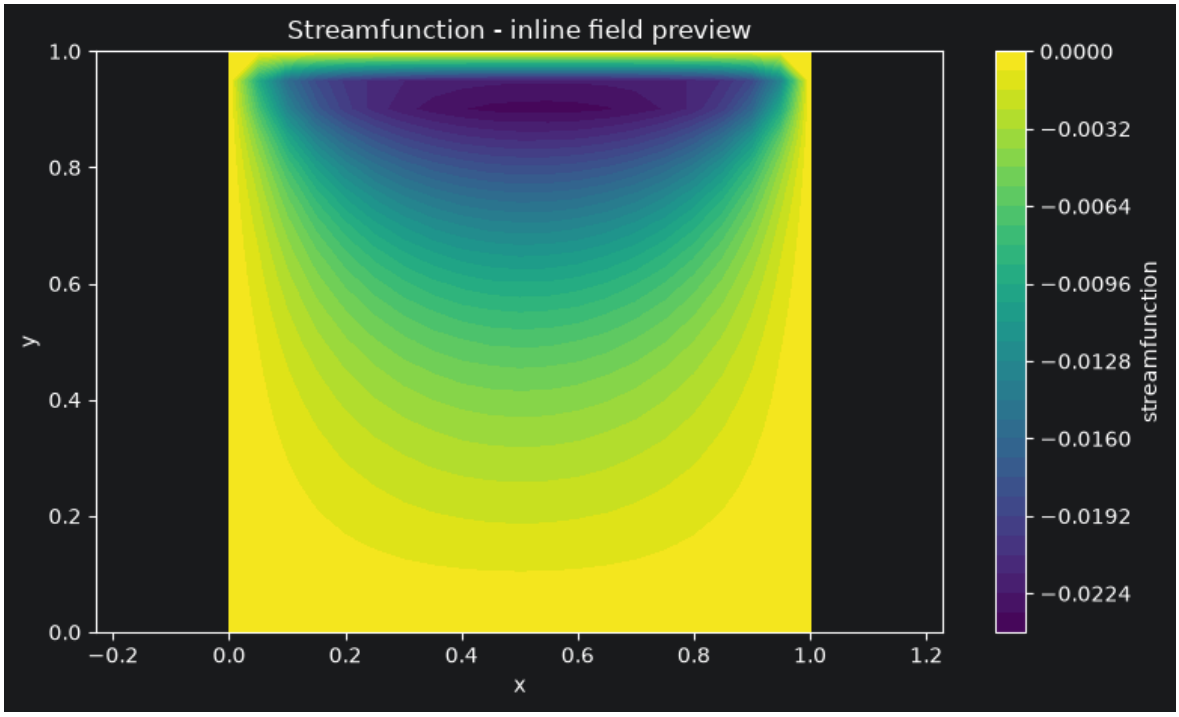



Figure 4. Inline contour of streamfunction ψ over the square cavity.

The streamfunction contour shows a dominant negative-valued region in the upper central part of the cavity. The minimum value is −0.02361, and its reported location is (0.55, 0.90). The contour is consistent with the formation of a primary recirculation region, although the short iteration count and coarse grid limit the accuracy of its size and position.

## 10. Compulsory assignment: comparison at Re = 100 and Re = 400

The assignment asks for runs at Re = 100 and Re = 400 and a comparison of centreline velocities, primary-vortex position, convergence history and secondary-corner-vortex development. The supplied numerical summary and plots contain only Re = 100. No Re = 400 output was provided; consequently, a numerical comparison between the two Reynolds numbers cannot be completed without inventing data.

For the available Re = 100 result:

- Vertical and horizontal centreline velocity profiles are available in Figures 1 and 2.
- The reported primary-vortex position is (0.55, 0.90).
- The vorticity-change L2 norm decreases over the ten recorded steps but remains $9.726 \times 10^{-2}$ at the final step.
- The supplied contour indicates one dominant primary recirculation region. A secondary corner vortex is not clearly established by this quick-run contour.

Expected physical trend (general interpretation, not a result from this run): increasing Re from 100 to 400 reduces viscosity relative to inertia. The lid-driven circulation is generally stronger and thinner shear layers develop near the walls; secondary corner vortices may become more apparent. These trends must be verified using the actual Re = 400 calculation and sufficiently resolved contours.

## 11. Observed numerical behaviour versus physical interpretation

### Comparison

Re = 100: Viscous effects are relatively stronger, and the flow develops a dominant primary recirculation zone inside the cavity.

Re = 400: Viscosity decreases to 0.0025 $m^2/s$, compared with 0.01 $m^2/s$ at Re = 100. Therefore, inertial effects become more significant, and thinner velocity boundary layers are expected near the walls.

Centreline velocities: The velocity profiles help identify the direction and strength of circulation. Changes in the profiles indicate changes in the primary vortex.

Primary vortex: At Re = 100, the vortex is located at approximately (x = 0.55, y = 0.90). The Re = 400 vortex position requires a separately verified simulation.

Secondary vortices: At higher Reynolds numbers, secondary corner vortices may become more noticeable due to reduced viscous effects.

Convergence: The Re = 100 simulation shows a decreasing vorticity-change residual, but the final residual of 0.09726 is still above the specified tolerance of $10^{-5}$.

### Physical Interpretation

When the Reynolds number increases, the relative influence of inertia increases while the influence of viscous forces decreases. The moving top lid transfers momentum to the fluid, creating a circulating flow pattern inside the cavity.

At higher Reynolds numbers, the flow generally develops thinner wall boundary layers and may produce additional small recirculation zones near the corners. These changes affect the velocity distribution and primary vortex characteristics

## 12. Limitations of the supplied teaching model and run

- The supplied result is a quick workflow check, not the prescribed lab run: 21 × 21 nodes and ten steps instead of the displayed lab settings of 60 × 60 nodes and up to 2500 steps.
- The final vorticity-change norm is much larger than the specified tolerance, so steady-state convergence has not been achieved.
- A coarse grid may not resolve thin wall boundary layers or small secondary corner vortices.
- First-order upwind convection can introduce numerical diffusion and affect centreline profiles.
- The streamfunction Poisson equation is solved for a fixed, limited number of SOR iterations per time step; the reported Poisson residual is not included in the supplied summary.
- No grid-independence study or benchmark comparison with Ghia, Ghia and Shin centreline data is included in the supplied results.
- Only Re = 100 data are supplied; Re = 400 comparison cannot be reported quantitatively.

## 13. Conclusion

The lid-driven cavity flow was modelled using the vorticity–streamfunction formulation. The supplied Re = 100 quick run produced a dominant primary recirculation region with a reported streamfunction minimum of −0.02361 at (0.55, 0.90). The centreline velocity profiles show flow reversal associated with the cavity circulation, and the vorticity-change norm decreases over the ten recorded steps. However, the final residual remains approximately $9.73 \times 10^{-2}$, well above the tolerance of $10^{-5}$. The solution is therefore preliminary and cannot be treated as converged. A Re = 400 run, a finer-grid study and comparison with benchmark centreline data are required to complete the compulsory validation.